

Classifying Tech News with Sentiment Analysis and Machine Learning

Krystofer Newman

**Intel Corporation &
Arizona State
University**

April 21st, 2019

Table of Contents

Summary	3
- Intellectual Merit	3
- Broader Impact	3
Introduction	4
- Reason for Project	4
- Project Objectives and Scope	4
- Brief Literature Review and Existing Designs	4
- Innovation and Uniqueness of Design	4
- Scope of Project	5
- Impact	5
- Organization of Proposal	5
Methodology	9
- Main Ideas of Developing Solutions	9
- Design Alternatives	10
- Concepts to be Used	11
Deliverable Consideration	11
Project Planning and Schedule	13
References	16
Appendix A	17

Summary

There are hundreds of websites out on the internet for news articles related to tech companies. However, major sites like TechCrunch and The Verge seem to lack any sentiment analysis even for how popular their platforms may be. With the rise of different applications of Artificial Intelligence and Machine Learning, this was a big surprise to me. Algorithmic processing and analysis are growing areas of research in the field of financial trading that aim to give investors insights into a particular company or stock without any human interaction or emotion assuming the algorithm was developed without errors. Applying this same concept of algorithmic processing to news articles using a sentiment analysis and classification algorithms to relevant articles may give readers and investors helpful insights to the impact of the event(s) on that particular company. It is important to apply this type of automated classification with as little human bias as possible to help readers view the article analysis without emotional response. Using a supervised learning algorithm and dataset from a small subset of pre-selected tech companies, we can provide readers personalized news and an unbiased analysis of whether that news event may impact the company in positive or negative ways. This automated classification aims to give readers insights into possible investments for companies in their portfolio and an overall trend of the news surrounding that particular company over a given range of time.

- Intellectual Merit

The research proposed in this project demonstrates how Machine Learning algorithms can be applied to classify news articles and give correlations of the news in a particular company's stock. This application provides insights into whether an article has positive, negative, or neutral impact to the company. Synonymous to other applications of Machine Learning in the Tech and Finance industries, this project provides a unique perspective into news around 5 major Technology and Software Companies.

- Broader Impact

Applying Machine Learning in creative ways to recognize patterns in data is a growing field of research. Research and Engineering professionals seek to unlock potential insights of their data. These insights can provide companies ways to optimize logistics, employee training, financial savings, and early recognition of genetic diseases. This project aims to give light to more creative analyses and inspire new research and applications of Artificial Intelligence.

In addition, this software application aims to de-mystify and educate on the integrations of Machine Learning into modern software development. All architectures and learnings will be shared so that others may develop their own applications. Encouraging engineers and data scientists to apply AI in unique ways is a large focus of the development in this project. At a very high level, all Artificial Intelligence is simply pattern recognition over a given dataset and using Linear Algebra combined with

Statistics. The patterns this project aims to identify are within news articles related to a variety of tech companies.

Introduction

- Reason for Project

Many popular news sites don't provide sentiment analysis to its readers. In an attempt to be unique and provide great value to tech news readers, we can provide a full platform for users to see how impactful news events are to a company and its possible correlation to the stock value. These insights may provide useful statistics of a company's recent trends in the news cycle and help a reader to decide whether it is a good time to invest or not. These suggestions are just that, this platform is not aimed to be a brokerage application telling users to invest or not, but simply providing helpful analysis of a particular company.

- Project Objectives and Scope

Deploy a full stack application to public domains that gives readers tech related news articles, sentiment analysis, and classification trends over selectable date ranges. The application will also provide varying stock values and charts over a given date range. The scope of the project will consist of architectural diagrams and specification requirements outlining the different frameworks and methodologies used in the development. Due to lack of time and resources, this project will be done using Agile in an iterative approach. The platform will provide readers with articles and correlation trends on five large tech companies: Google, Amazon, Facebook, Intel, and Microsoft.

- Brief Literature Review and Existing Designs

Out on the web there seems to be a plethora of engineers applying machine learning algorithms to analyze political data on social media platforms such as Twitter and Facebook [1][5]. This research aims to apply sentiment analysis algorithms to text and articles posted by elected officials to gain a deeper insight and analysis of the overall trend of their posts. There are also similar designs focusing on financial trading data and applying algorithmic processing to publicly traded stock information [2]. Using a variation of these applications, we can analyze the sentiments in different news articles on particular companies and provide a holistic approach to a company's overall news trend and whether that company has a positive or negative correlation in the news.

- Innovation and Uniqueness of Design

As previously mentioned, there are other similar applications using supervised learning algorithms and sentiment analysis, however through research it seems there are few (if any) platforms providing readers sentimental analysis on tech related news. As an engineer in the tech company, I do my best to follow tech news through

different platforms such as TechCrunch and The Verge, however if I wanted to see the trends of a particular company and how they were viewed in the news I would have to manually dig through articles over time. With an automated approach, I can quickly read articles on these companies while having an overall trend of both the sentimental analysis and stock values.

- **Scope of Project**

The scope of the project will be a deployed full stack web application including a User Interface, Service Layer, Connected Database, and various python scripts running on daily jobs that pull article data from open source APIs as well as classifications through machine learning algorithms. The overall architecture, technical specification documents, and development schedule will be provided to give progress reports during the milestones of the course.

- **Impact**

As aforementioned, this project aims to encourage engineers to integrate Artificial Intelligence to modern, open-source software applications. This project also gives light to how a simple three tier application can be quickly developed and deployed. The impact of the project may be measured by the number of software engineers that find the readings helpful as a training guide or reference for the ease of implementation.

- **Organization of Proposal**

The remainder of the proposal will cover the referenced literature for this project and the preliminary work completed. The work completed thus far includes architectural designs, data models, chosen frameworks, hosting solutions, and decisions involved. The proposal will also explore existing designs and compare the implementations and designs of this project showing scope and potential impacts of the work.

The proposal will also cover Methodologies used in this project. One main focus of the implementations in this project is reusability. As the pages of the User Interface are all very similar in structure, we can design the code to be reusable and lessen the technical footprint and size of the interface. This approach also helps developers in debugging and testing phases of development.

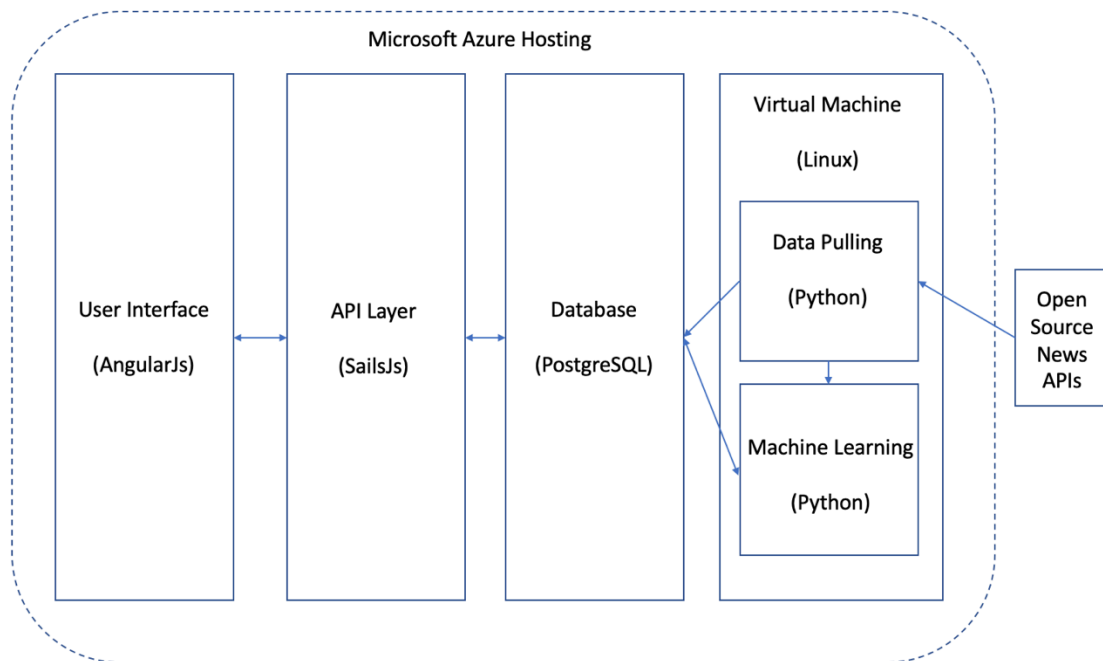
Project Deliverables and Schedule will also show the progress and implementations over the course of the semester. Each deliverable was split into small achievable and realistic goals. A vast majority of the preliminary tasks completed in the project were related to setup and configurations of the different elements such as the Database and API layers. These deliverables follow the project plan very closely and are managed in the schedule as well as the budget and justifications.

The Budget and Justifications section of the proposal will explain costs of development and hosting solutions for the project. Development tools and equipment used throughout the semester will also be explained and validated.

Finally, all references will be listed as well as appendices for the project proposal. As many of the components and frameworks are well documented through Google and Microsoft's developer programs, these links and references provide useful connection information and examples of different operations for their respective components.

Literature Review and Related Work

The preliminary work done for this project includes all setup and configuration of each layer of the architecture shown below.



The initial scaffolding of the User Interface and API layer were also completed within the first deliverable. In addition, configurations and connections were established between each component of the architecture. These configurations required many trials and errors as documented versions varied between current version of the modules.

As this project aims to analyze news articles from 5 different tech companies, there are a variety of python scripts that were developed and hosted within the Linux Virtual Machine. These scripts pull data from EventRegistry [*] and a Google News API [*]. These scripts are also run on jobs that execute 3 times a day to always provide Users with relevant information and ratings.

The following will show code examples from both the Python and UI elements of the

project that is reused across that parent entities layer.

An example of one of the python scripts pulling data is shown below:

```
# read the microsoft data..
response = requests.get('https://newsapi.org/v2/everything?q=microsoft&apiKey=')
json_data = json.loads(response.text)

for article in json_data["articles"]:
    if article:
        try:
            source = article["source"]["name"]
            author = article["author"]
            title = article["title"]
            desc = article["description"]
            url = article["url"]
            img = article["urlToImage"]
            pub = article["publishedAt"]
            content = article["content"]
            rank = None

            # Insert some data into table
            try:
                cursor.execute("INSERT INTO microsoft (source, author, title, desc, url, img, pub, content, rank) VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s)" % (source, author, title, desc, url, img, pub, content, rank))
                print("inserted into microsoft Table!")
            except Exception:
                print("Error inserting to db, duplicate")

        except KeyError:
            print("something went wrong")
```

An example of a portion of reusable code in the User Interface is shown below:

```

angular
  .module('starterAngularFractal')
  .directive('feedCard', feedCard);

/* @ngInject */
function feedCard() {
  var directive = {
    bindToController: true,
    controller: FeedCardController,
    controllerAs: 'vm',
    link: link,
    restrict: 'E',
    replace: true,
    scope: {
      newsInfo: '=',
    },
    templateUrl: 'common/directives/news/news.template.html'
  };
  return directive;

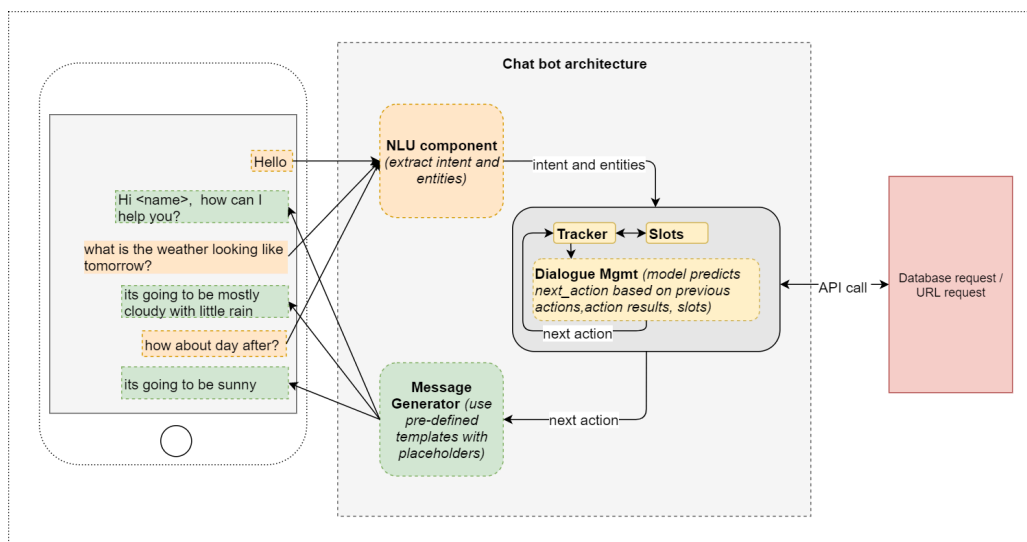
  function link(scope, element, attrs) {
  }
}

/* @ngInject */
function FeedCardController() {

  var vm = this;

```

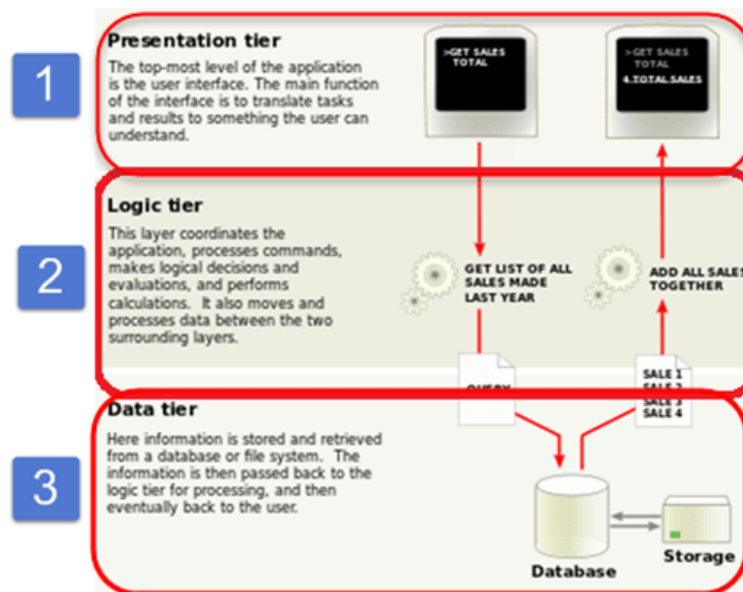
Similar designs to this project include AI Chatbot Applications and other Financial Tools like ETrade. One example shows how a similar architecture implements AI for a Chatbot Tool below [6]:



The designs are very similar and shows how applications can integrate AI with simple 3 or 4 tier architecture designs.

Other examples and generalizations can be shown like the one listed below. This design does not include any interfaces for AI but similarities show that simple 3-4 tier

architectures can be quickly developed and deployed with many advantages.



The architecture for this project extends the common 3 tier architecture to a 4 tier, adding a Virtual Machine that hosts the Python elements of the project. These elements include automated scripts that query open source APIs for data on each of the 5 companies, format the data and insert it into the database. In addition, the Virtual Machine also hosts the Machine Learning model that ranks the articles using Supervised Learning.

Methodology

- Main Ideas of Developing Solutions

Throughout the course of my professional engineering career and education, I have been involved with a variety of projects and applications. Most of my professional work has been geared towards full stack applications that support Intel's internal processes in Supply Chain. After roughly 5 years of experience in the industry, I have seen and implemented a wide range of designs and architectures while gaining knowledge of large-scale enterprise applications and how they are supported. Along with my knowledge from my undergraduate degree and graduate degree as well as my passion for web development and newly gained skills in machine learning, I wanted to pursue a similar type of design for my Applied Project on a smaller scale given the courses timeframe.

Our development team follows Agile methodologies as well as Scrum to ensure our projects are successful. We have roughly 10 team members including a project manager and scrum master that coordinate our tasks and timelines of development. Most of our applications are developed in .Net and C# code that interface with Sql Databases. For this Applied Project, the methodology applied will be a variant of Agile due to timelines and resources. Due to the limited resources being myself on the project, I will implement a varied version of the Agile methodology and use the Milestones and update reports of the course as Scrum updates. This variation will clearly define broken down tasks (as covered in CSE566 Software Project

Management).

My ideas for this project stemmed from a small proof of concept project I volunteered on at work which applied machine learning to determine if news events may be impactful to Intel's Supply Chain. Companies like DHL use a similar platform to predict impactful events in their logistics [3]. Expanding on these concepts and the previously mentioned sentimental analyses done on Social Media [5] platforms, having useful insights into news articles related to different tech companies in my personal investment portfolio was a very promising idea.

As mentioned before, my current team supports .Net and Sql Databases however, most of my experience lies in open source packages and frameworks like AngularJs, NodeJs, and various python tools. I prefer the open source model of development due to the inherent separation of concerns. In AngularJs version 1.x, build and compile packages are installed via Node packages and packages used in the User Interface are installed via Bower. In my opinion, this allows a clear visualization of the structure of the project and a clearer path when debugging or supporting the application. In contrast, newer versions of Angular and even .Net use all installed packages in a combined manner through installers like Node and Nuget, respectively.

Most of the ideas I had for this project stemmed from the aforementioned applications I have researched where engineers are analyzing data from Social Media sites like Twitter, Facebook, and Reddit. I felt putting a unique spin on the existing applications would give users a wider range of information on a particular company all in one place, including updated news articles and stock information. The 5 companies listed above are all companies I currently invest in as a part of my retirement portfolio, and as I did a routine check of the health of my 401k investments, I began to think of how useful it would be to have an aggregated platform to see the health of the companies I invest in, including sentimental analysis.

- Design Alternatives

The general architecture of the application is more or less the same across my design alternatives with variations in the frameworks used to build the individual components. Most applications are generally 3 tiered with a User Interface, Service Layer, and Database. Given my experience in the industry and personal preferences, I plan to use Javascript and Python as the main languages for this application because it is a web-based application. Listed below are a few design alternatives I considered from a technical standpoint.

.Net Alternative

I originally considered building the application using .Net and C# due to the popularity and because it is a main technology studied in the CSE 598 courses Distributed Software Development and Software Integration and Engineering. This design would consist of using ASP .Net as a UI and various classes and services developed in C# to interact with a Sql Database. From my research it also seemed very reasonable to host and deploy these types of applications on the Microsoft Azure platform.

React Alternative

My second design alternative was aimed at stepping out of my comfort zone of development and targeting the open source framework from Facebook, React. Many large tech companies are moving to React and it is quickly gaining popularity in the web development world. I took a React training course and it seemed very similar to newer versions of Angular and I thought this might be an opportunity to enhance those skills. Given the short timeframe of the course and limited bandwidth of my personal time, I felt learning this newer framework might not be a viable option.

Angular Alternative

My final design alternative was to use the open source framework from Google, AngularJs. Many of the projects I have been involved with at Intel use Angular in the UI component as that is our company guidance for security and maintainability reasons. With my wide range of experience and knowledge using the framework, and the limited timeframe of the semester, I opted for this design using Javascript in both the UI and Service Layer. This design uses some of the latest technology in the industry minus the version of AngularJs (1.x). The service layer is another open sourced framework we use at Intel called SailsJs which is built on top of NodeJs. The ease of use for creating models and RESTful endpoints give this framework preference. The database to be used in this design is PostgreSQL as there is great connectivity options between SailsJs and PostgreSQL set in place by the open source community.

- Concepts to be Used

As I finish my Master's Degree, I wanted to give this project a culminative approach of many of the concepts learned over the course of my experience in the engineering program. Fortunately, I also use many of the concepts and methodologies discussed in the program in my professional career at Intel. One of the main concepts is reusability. The AngularJs framework lets developers leverage reusability through components like Factories and Directives. These components allow developers to decide what models inherit these functionalities to write less code and in turn, debug less code in maintenance. Another concept to be used will be Object Oriented Programming. PostgreSQL stores data in a Sql like structure which simplifies the end to end data storing techniques where the Service Layer will read and write similar object structures. A simple example of a news article object may include the attributes or fields as: Title, Author, URL, Source, Date, etc. Using open source APIs to pull news articles, [4] the structure of the data is very clean and uniform across all queries.

Deliverable Consideration

- Software
 - The full stack application will consist of varied 4 tier architecture including a User Interface, Service Layer, Database, and various python scripts for a machine learning model and a data processing pipeline hosted in a Virtual

Machine.

- System
 - The application will be hosted through a platform (to be decided) such as Amazon Web Services or Microsoft Azure. This hosting system should support Virtual Machines that will enable the hosting of each individual components like the UI and Database. Another implication and deciding factor of the hosting platform and the VM is the reliability and security of the platform. Many open source APIs are free but require special keys for individual users which should be secured by the system.
- Publications
 - Designs and documentation will be hosted through my personal website (krysnewman.com) for visibility and ease of use for both the staff of the course and the developer (myself) of the system. All progress reports will be submitted through both Canvas and the personal site.
- Reusable design
 - Various code and models will be reused as seen fit during development. As mentioned above in the Concepts to be Used, this is a main aspect of the code design for the project.

An example of one of the extendable models for each news item is listed below. This model fits each tech company news article and data is formatted to fit this accordingly:

```
attributes: {  
  id: {  
    type: 'number',  
    unique: true,  
    autoIncrement: true  
  },  
  source: {  
    type: 'string',  
    required: true  
  },  
  author: {  
    type: 'string',  
    required: true  
  },  
  title: {  
    type: 'string',  
    required: true,  
    unique: true  
  },  
  description: {  
    type: 'string',  
    required: true  
  },  
  url: {  
    type: 'string',  
    required: true  
  },  
  urltoimage: {  
    type: 'string',  
    required: true  
  },  
  publishedat: {  
    type: 'ref',  
    required: true ,  
    columnType: 'datetime'  
  },  
  content: {  
    type: 'string',  
    required: true  
  },  
  rank: {  
    type: 'number'  
  },  
  confidence: {  
    type: 'number'  
  },  
  mlline: {  
    type: 'string'  
  }  
}  
};
```

Project Planning and Schedule

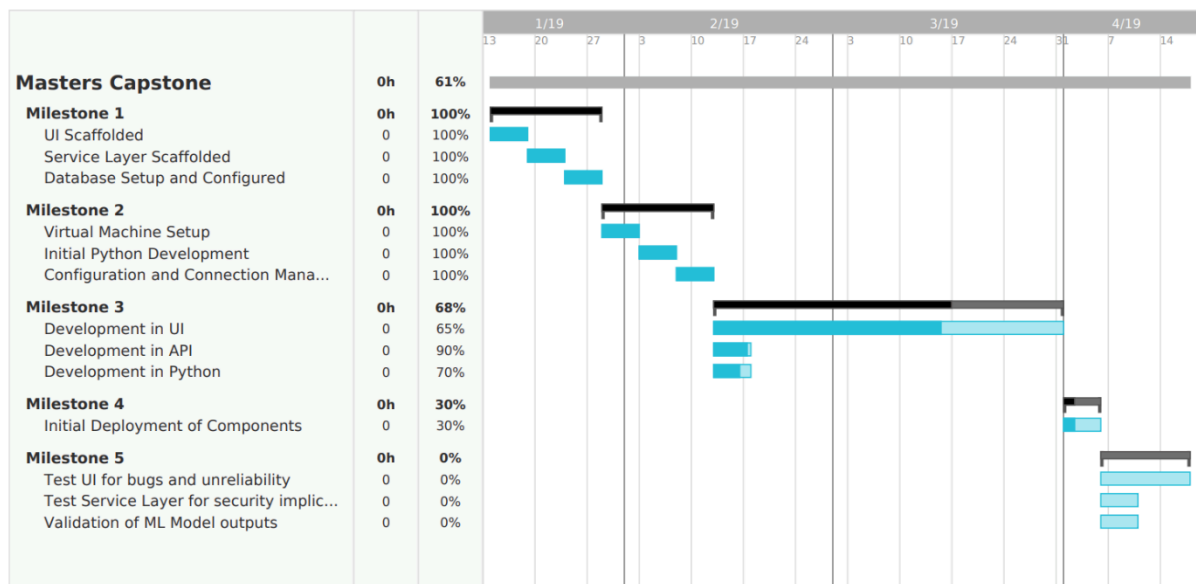
- Plan
 - Using an Agile-like approach to the project would be very beneficial in breaking up the tasks of development. Many of the components and service APIs in the service layer may have to be developed in tandem for verification and validation of data flow and correctness.
 - The machine learning model may be one of the last components to be deployed for iterative development and validation of the data as it is collected by the system.

- Schedule

The schedule for the project is shown below in a Gannt Chart and is broken down into achievable tasks for each milestone

Using TeamGantt [11], we can quickly visualize the different dependencies for the


Created with Free Edition



project and give an estimate for each Milestone. As shown, development of the project extends from February to the end of March with room for deployment and testing before the conclusion of the semester and Final Report.

As shown above, the project is roughly 50% complete and still requires development in both the User interface and Python elements to create the Machine Learning model that classifies the news articles.

These components will then need to be deployed to the Azure hosting platform and validated that they connect to their dependencies correctly, i.e. the UI to Service Layer and so on.

- Explain the milestones
 - Milestone 1 / Report 2: Initial Project Creation
 - UI Scaffolded
 - Service Layer Scaffolded
 - Python script development
 - Milestone 2 / Report 3: Hosting
 - Virtual Machine created on hosting platform
 - Necessary packages installed on VM
 - Db created and setup
 - Milestone 3 / Report 4: Continued Development
 - Development in UI
 - Development in Service Layer
 - Python development for data pipeline and ML model
 - Milestone 4 / Report 5: Deployment and Validation
 - Initial deployment of components
 - Milestone 5 / Report 6: Verification and Validation of System
 - Test UI for bugs and unreliability
 - Test service layer for security implications and private endpoints
 - Validation of Machine Learning model outputs and classifications
 - Milestone 6 / Final Report: Submit Findings and Results

Budget and Justification

- **Personal Cost**

The costs of this project have been very minimal over the course of its implementation. As all of the frameworks and news sources are open source and free, any developer can begin using them without costs.

Any costs associated with the project stem from the hosting platform, Microsoft Azure. The platform provides a free trial subscription of \$200 for any new customer to use. The estimated costs of all running services on the Azure platform over a month's period was ~\$90. As the dataset is relatively small for this application, roughly 700 rows of data per Tech Company, as well as multiple queries from the UI to the Service Layer API, these costs are very minimal.

- **Equipment**

There is no associated physical equipment for this project as it is all Software hosted through the Azure Cloud platform. Microsoft most likely has distributed servers that connect to its Azure platform however I did not have access to the visibility of these.

- **Tools and Justifications**

The tools used for this project include the free IDE Visual Studio Code [9]. This IDE is very developer friendly and has a growing ecosystem of plugins that extend the functionalities of the program.

I chose this IDE as I am already familiar with the platform and its extensions. I use this tool daily within my professional career. The other driving factor behind this tool is the cost, Visual Studio Code is free. In contrast, other IDEs require a license such as WebStorm or Sublime Text.

References

- [1] Singh, M. (n.d.). Trump and the media. Retrieved January 24, 2019, from <https://amiham-singh.github.io/>
- [2] Yip, J., & Yip, J. (2018, November 25). Algorithmic Trading using Sentiment Analysis on News Articles. Retrieved from <https://towardsdatascience.com/https-towardsdatascience-com-algorithmic-trading-using-sentiment-analysis-on-news-articles-83db77966704>
- [3] DHL: Cracking Logistics with the help of Machine Learning. (2018, November 13). Retrieved January 24, 2019, from <https://rctom.hbs.org/submission/dhl-cracking-logistics-with-the-help-of-machine-learning/>
- [4] <https://newsapi.org/s/google-news-api>
- [5] Geitgey, A., & Geitgey, A. (2018, August 15). Text Classification is Your New Secret Weapon – Adam Geitgey – Medium. Retrieved January 21, 2019, from <https://medium.com/@ageitgey/text-classification-is-your-new-secret-weapon-7ca4fad15788>
- [6] Kompella, R., & Kompella, R. (2018, February 09). Conversational AI chat-bot - Architecture overview. Retrieved from <https://towardsdatascience.com/architecture-overview-of-a-conversational-ai-chat-bot-4ef3dfef52e>
- [7] <https://github.com/EventRegistry/event-registry-python/wiki/Searching-for-events>
- [8] CHEN, Y. (2017). SERVICE-ORIENTED COMPUTING AND SYSTEM INTEGRATION: Software iot big data and ai as services. KENDALL HUNT.
- [9] <https://code.visualstudio.com/>
- [10]

<https://getdata.io/data-sources/4779-venturebeat-tech-news-that-matters-news-headlines>

[11] <https://www.teamgantt.com/>

Appendix A

Code Samples

```
(function () {
    'use strict';

    angular
        .module('starterAngularFractal')
        .directive('feedCard', feedCard);

    /* @ngInject */
    function feedCard() {
        var directive = {
            bindToController: true,
            controller: FeedCardController,
            controllerAs: 'vm',
            link: link,
            restrict: 'E',
            replace: true,
            scope: {
                newsInfo: '=',
            },
            templateUrl: 'common/directives/news/news.template.html'
        };
        return directive;

        function link(scope, element, attrs) {

        }
    }
})
```

```
/* @ngInject */
function FeedCardController() {

    var vm = this;

    vm.rankPopover = createRankPopover(vm.rank);

    function createRankPopover(rank) {
        switch(rank){
            case 0:
                vm.rankPopover = "We ranked this article as having high negative impact";
                break;
            case 1:
                vm.rankPopover = "We ranked this article as having low negative impact";
                break;
            case 2:
                vm.rankPopover = "We ranked this article as not relevant or no impact";
                break;
            case 3:
                vm.rankPopover = "We ranked this article as having low positive impact";
                break;
            case 4:
                vm.rankPopover = "We ranked this article as having high positive impact";
                break;
        }

        return vm.rankPopover;
    }
}
```

```

# Construct connection string
conn_string = "host={0} user={1} dbname={2} password={3} sslmode={4}".format(host, user, dbname, password, sslmode)
conn = psycopg2.connect(conn_string)
print("Connection established")

cursor = conn.cursor()

er = EventRegistry(apiKey = "a13a374e-cd84-43a0-9e0f-85df2a79fb58")
q = QueryArticlesIter(
    keywords = "FB",
    lang = "eng"
)

res = q.executeQuery(er, maxItems = 500, sortBy = "rel")

for article in res:
    #print(article)
    try:
        source = article["source"]["title"]
        author = ""
        title = article["title"]
        desc = article["description"]
        url = article["url"]
        img = article["image"]
        pub = article["date"]
        content = article["body"]
        rank = None

        # Insert some data into table
        try:
            cursor.execute("INSERT INTO facebook (source, author, title, description, url, urltoimage, rank) VALUES (%s, %s, %s, %s, %s, %s, %s)" % (source, author, title, desc, url, img, rank))
            print("inserted into facebook Table!")
            conn.commit()

        except psycopg2.DatabaseError as error:
            print(error)
            conn.rollback()
            continue

    except Exception as e:
        print(e)
        pass

```